CprE/SE 4910 WEEKLY REPORT 05

10/21/2026 – 10/27/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

The group discussed with both the client and a TA for the course. The team gained a significant amount of information regarding the scope of the project as well as the way the design document should be set up. The group has been given more unique parts that each member needs to work on for the project.

Past Week Accomplishments

- Group Work:
  - Discussed with Client and TA.
- Nakshatra Gupta:
  - Contributed to the team discussion on splitting the project into smaller milestones, drawing on what the extraction script can and cannot read from the sample plans. Assigned as the main person to handle the JSON schema that connects both computer vision and graphics.
  - Shared the first version of the JSON schema with the team. The schema defines how extracted plan data (stories, walls, rooms, openings, roof, and dimensions) is passed from the computer vision side to the graphics side. It was built from one of the client's sample plans, so it covers the features of that plan only.

```
{
  "building_id": "sample_001",
  "source_file": "sample_floorplan_01.pdf",
  "stories": [
    {
      "story_index": 0,
      "base_height_m": 2.7,
      "walls": [
        {"id": "w1", "start": [0.0, 0.0], "end": [6.8, 0.0], "thickness_m": 0.2, "exterior": true},
        {"id": "w2", "start": [6.8, 0.0], "end": [6.8, 11.0], "thickness_m": 0.2, "exterior": true},
        {"id": "w3", "start": [6.8, 11.0], "end": [0.0, 11.0], "thickness_m": 0.2, "exterior": true},
        {"id": "w4", "start": [0.0, 11.0], "end": [0.0, 0.0], "thickness_m": 0.2, "exterior": true},
        {"id": "w5", "start": [3.4, 0.0], "end": [3.4, 11.0], "thickness_m": 0.1, "exterior": false}
      ],
      "rooms": [
        {"id": "r1", "name": "Bedroom 1", "boundary_wall_ids": ["w1", "w5", "w3", "w4"], "area_m2": 18.6},
        {"id": "r2", "name": "Bedroom 2", "boundary_wall_ids": ["w5", "w2", "w3"], "area_m2": 21.9}
      ],
      "openings": [
        {"id": "d1", "type": "door", "wall_id": "w4", "position_m": 5.2, "width_m": 0.9},
        {"id": "win1", "type": "window", "wall_id": "w1", "position_m": 1.5, "width_m": 1.2},
        {"id": "win2", "type": "window", "wall_id": "w2", "position_m": 6.0, "width_m": 1.2}
      ]
    }
  ],
  "roof": {
    "type": "hip",
    "pitch_deg": 30
  }
}
```


- Rishiraj Parmar:
    - Reviewed the shared JSON schema and documented additional fields needed for 3D reconstruction.
- Ryan McNab:
    - After scaling back the project with the Buildertrend contact, the team assigned the role of UI design and prototyping to Ryan McNab.
    - As a first step of the improved prototype, the current prototype computer vision model, which cannot be trained, was replaced with YOLOv11 and the training has started.
    - Using a generative AI model, 600+ new 2d models were generated will be trained upon in the future.
    - Started training the YOLO model on 20 epochs.
- Daniel Hargrave:
    - Looked into YOLO more now that Buildertrend approved its use.
        - Set up some of the initial environment for it and learned some limitations for it.
        - Looked at the prototype and tried to get it switched over to YOLO but ran into several issues.
        - Found out just how much images that YOLO would need in order to work.
            - Looked back into the research papers that were read before this. Found one publicly available dataset that would be helpful but can’t be used for commercial purposes. Might try and use it anyways at least temporarily in order to get the model working.
- Jase Beaubien:

    - Planned and researched graphics.
    - Coordinated high-level plans.

Pending issues

- Nakshatra Gupta:
    - The schema was built from a single blueprint, so it will likely need changes to cover features in the other plans, such as different roof shapes, porches, and different floor layouts. The extraction script works on all four sample plans, but it is not known whether future plans will be formatted differently.
- Rishiraj Parmar: N/A
- Ryan McNab:
    - High numbers of epochs take a long time to run. For example 120 epochs, which is what the research provided would take over 17 hours. This may still work if ran overnight, but reducing the number may be the only option.
- Daniel Hargrave:
    - It would probably work to use the dataset that was found for now in the prototype, but it likely can't be used in the final product. Might have to find a publicly available free dataset.
- Jase Beaubien:
    - Don't have models to work with.

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Shared the first version of the JSON schema with the team and contributed to the milestone and scope discussion.	3	22
Rishiraj Parmar	Reviewed the shared JSON schema and identified additional fields needed for 3D reconstruction.	3	23
Ryan McNab	Time was spent mostly replacing the computer vision model with the selected one and training it.	3	28

Daniel Hargrave	Studied the documents provided and tried to separate screenshots into different categories based on what each does. Attempted to come up with some basic plans for how to move forward given the new information.	3	23
Jase Beaubien	Planning and research for graphics.	3	21

Comments and Extended Discussion

This week the group learned a lot from the client.

Plans for the upcoming week

- Nakshatra Gupta:
  - Collect feedback on the schema from the Computer Vision and Graphics teams. Revise it to cover features from the other three sample plans and check it against Rishiraj's integration prototype fields.
- Rishiraj Parmar:
  - Work on calculating floor base elevations from story heights to support correct floor placement in the 3D reconstruction prototype.
- Ryan McNab:
  - Train the model on a larger dataset and more epochs.
  - In the future, will be in charge of UI design and prototyping, while helping improve the YOLO model.
- Daniel Hargrave:
  - For the future, will be in charge of getting our project to work with the YOLO model.
  - Implement the YOLO model fully with the temporary dataset. Will try and put this in the prototype.
- Jase Beaubien:
  - Simulate basic models in the program.